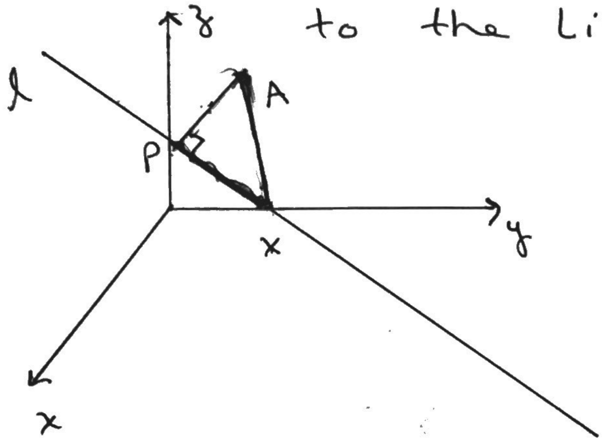
Distance Between a Point and a Line

Example: Find the distance from point $A(1,2,3)$ to the line $\vec{r} = [0,1,5] + t[3,4,1]$



1. pick a point on the line, l.
2. find the magnitude of the projection of $\vec{AX}$ onto l. (ie: $|\vec{PX}|$)
3. using Pythagorean theorem with hypothenuse length $|\vec{AX}|$, find the distance of $\vec{AP}$.

$\rightarrow$ check if $A(1,2,3)$ is on the line:

$$\left.\begin{array}{l} 1 = 3t \longrightarrow t = \frac{1}{3} \\ 2 = 1 + 4t \longrightarrow t = \frac{1}{4} \\ 3 = 5 + t \longrightarrow t = -2 \end{array}\right\} \Rightarrow A(1,2,3) \text{ is } \underline{\text{not}} \text{ on the line, } l.$$

$\rightarrow$ pick a point on the line, l.

use $(0,1,5)$

→ find length of $\vec{PX}$ using proj of $\vec{AX}$ onto $\vec{d}$.

$$|\vec{PX}| = |\text{proj}_{\vec{d}}\, \vec{AX}| = \frac{|\vec{AX} \cdot \vec{d}|}{|\vec{d}|}$$

$$= \frac{|[-1,-1,2] \cdot [3,4,1]|}{\sqrt{9+16+1}}$$

$$= \frac{|-3-4+2|}{\sqrt{26}}$$

$$= \frac{5}{\sqrt{26}}$$

→ Length of $\vec{AX} = \sqrt{(-1)^2+(-1)^2+(2)^2}$

$$= \sqrt{6}$$

→ By Pythagoras,

$$(AP)^2 + (PX)^2 = (AX)^2$$

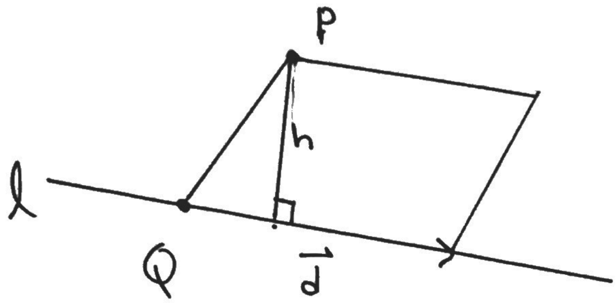
$$(AP)^2 = (\sqrt{6})^2 - \left(\frac{5}{\sqrt{26}}\right)^2$$

$$(AP)^2 = 6 - \frac{25}{26}$$

$$AP \doteq 2.24 \text{ units.}$$

Shortest distance from a point to a line.

Method 2



$$\text{Area of Parallelogram} = |\vec{PQ} \times \vec{d}|$$

$$\text{Area of Parallelogram} = h \cdot |\vec{d}|$$

h is the shortest distance

$$\therefore h = \frac{\text{Area}}{|\vec{d}|}$$

$$h = \frac{|\vec{PQ} \times \vec{d}|}{|\vec{d}|}$$

Distance Between 2 Lines

NOTES: ① The distance between 2 non-skew lines is 0. (ie: they intersect)

② The shortest distance between skew lines is the length of the common $\perp$ line.

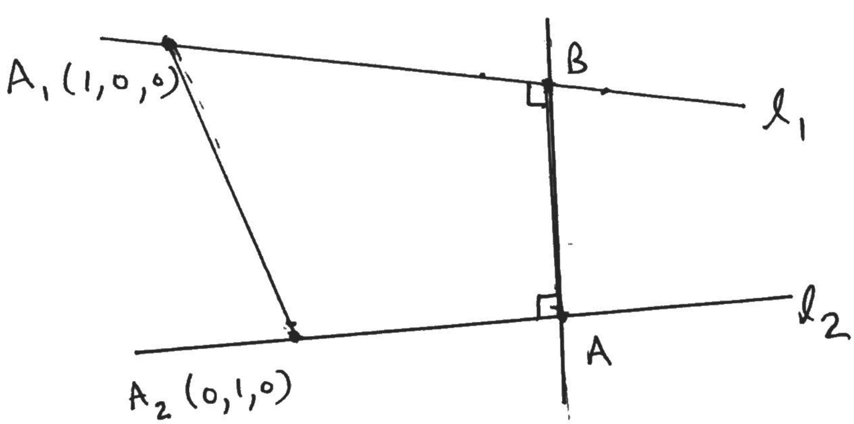
Example: Determine the distance between the 2 skew lines

$$l_1: \; x-1 = y = z \quad \underline{\text{and}} \quad l_2: \; \frac{x}{2} = \frac{y-1}{-1} = z$$

① Find the direction vectors and points on the lines.

$\vec{d_1} = [1, 1, 1]$; $\vec{d_2} = [2, -1, 1]$

$A_1 = (1, 0, 0)$; $A_2 = (0, 1, 0)$



If we project vector $\overrightarrow{A_2A_1}$ onto vector $\overrightarrow{AB}$ and take the magnitude, then that yields the required distance.

$$\therefore \overrightarrow{A_2A_1} = [-1, 1, 0]$$

$$\vec{n} = \vec{d_1} \times \vec{d_2}$$

$$\begin{array}{cccccc} 1 & 1 & 1 & 1 & 1 & 1 \\ 2 & -1 & 1 & 2 & -1 & 1 \end{array}$$

$$\vec{n} = [2, 1, -3]$$

$$\therefore |\overrightarrow{AB}| = \left| proj_{\vec{n}} \overrightarrow{A_2A_1} \right|$$

$$|\overrightarrow{AB}| = \frac{|\overrightarrow{A_2A_1} \cdot \vec{n}|}{|\vec{n}|}$$

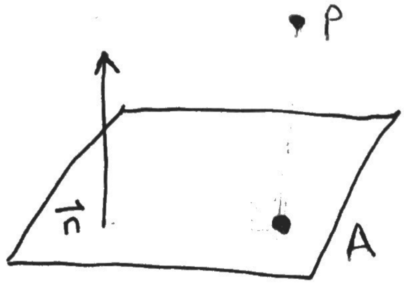
$$|\overrightarrow{AB}| = \left| \frac{[-1, 1, 0] \cdot [2, 1, -3]}{\sqrt{(2)^2 + (1)^2 + (-3)^2}} \right|$$

$$|\overrightarrow{AB}| = \frac{1}{\sqrt{14}}$$

$$|\overrightarrow{AB}| \doteq 0.27 \text{ units}$$

Distance from a Point to a Plane

ex: Find the distance from the point $P(-2, 5, 8)$ to the plane $2x + 3y - 7z + 5 = 0$



$\vec{n} = [2, 3, -7]$

let $x = -1$; $y = -1$,

$\Rightarrow 2(-1) + 3(-1) - 7z + 5 = 0$

$\therefore z = 0$

$\therefore$ The pt. on the plane is $A(-1, -1, 0)$

$\vec{AP} = [-1, 6, 8]$

$$\therefore \left|proj_{\vec{n}} \vec{AP}\right| = \frac{|\vec{AP} \cdot \vec{n}|}{|\vec{n}|}$$

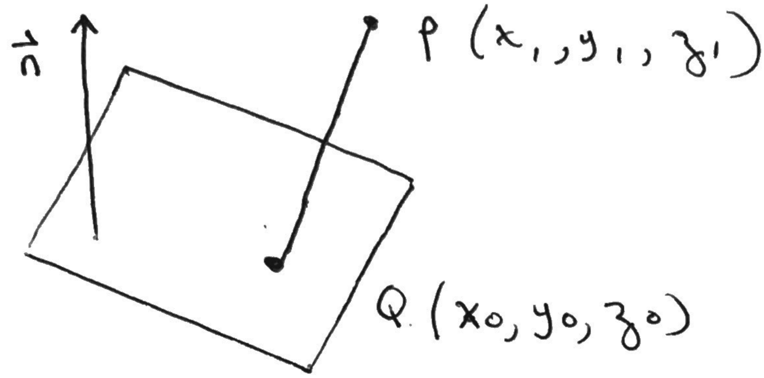
$$= \frac{|[-1, 6, 8] \cdot [2, 3, -7]|}{\sqrt{4 + 9 + 49}}$$

$$= \frac{|-2 + 18 - 56|}{\sqrt{62}}$$

$$= \frac{40}{\sqrt{62}}$$

$\doteq 5.08$ units

Distance from a point to a Plane.



$$\vec{QP} = [x_1 - x_0, y_1 - y_0, z_1 - z_0]$$

Equation of Plane is $Ax + By + Cz + D = 0$

point $Q(x_0, y_0, z_0)$ is on the plane.

$$\therefore \quad A(x_0) + B(y_0) + C(z_0) + D = 0$$

$$\Rightarrow \quad D = -Ax_0 - By_0 - Cz_0$$

$$\text{distance} = |\text{proj}_{\vec{n}} \vec{QP}|$$

$$\text{distance} = \frac{|\vec{QP} \cdot \vec{n}|}{|\vec{n}|}$$

$$\text{distance} = \frac{[x_1 - x_0, y_1 - y_0, z_1 - z_0] \cdot [A, B, C]}{\sqrt{A^2 + B^2 + C^2}}$$

$$\text{distance} = \frac{|A(x_1 - x_0) + B(y_1 - y_0) + C(z_1 - z_0)|}{\sqrt{A^2 + B^2 + C^2}}$$

$$\text{distance} = \frac{|Ax_1 - Ax_0 + By_1 - By_0 + Cz_1 - Cz_0|}{\sqrt{A^2+B^2+C^2}}$$

$$\text{distance} = \frac{|Ax_1 + By_1 + Cz_1 - Ax_0 - By_0 - Cz_0|}{\sqrt{A^2+B^2+C^2}}$$

$$\text{distance} = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2+B^2+C^2}}$$

Distance Between a Line & a plane.

ex: find the distance from line

$$l: \vec{r} = [2,0,1] + t[1,4,1]$$

$$\pi: 2x - y + 2z - 4 = 0$$

$\therefore$ parallel.

Solution: note: $\vec{d} \cdot \vec{n} = [1,4,1] \cdot [2,-1,2] = 0$

$$d = \frac{|\vec{n} \cdot \vec{PQ}|}{|\vec{n}|}$$

$\vec{n} = [2,-1,2]$

$P = (2,0,1)$

Q: Let $x=1$, $y=1$

$$2(1) - (1) + 2z - 4 = 0$$

$$2z = 3$$

$$z = \frac{3}{2}$$

$\therefore Q = (1, 1, \frac{3}{2})$

$$\vec{PQ} = (1,1,\tfrac{3}{2}) - (2,0,1)$$

$$= [-1, 1, \tfrac{1}{2}]$$

$$d = \frac{|[2,-1,2] \cdot [-1,1,\frac{1}{2}]|}{\sqrt{(2)^2 + (-1)^2 + (2)^2}}$$

$$d = \frac{|-2-1+1|}{\sqrt{4+1+4}}$$

$$d = \frac{2}{\sqrt{9}}$$

$$d = \frac{2}{3} \text{ units.}$$

Distance Between 2 Planes

ex: Find the distance between the planes

$\pi_1: \ 2x - y - z - 1 = 0$

$\pi_2: \ 2x - y - z - 4 = 0$

Solution: note: $\vec{n}_1 = \vec{n}_2 \Rightarrow$ parallel

Find a point on π_1: Let x and $y = 0$

$\Rightarrow z = -1$

$\therefore P = (0, 0, -1)$

Find a point on π_2: Let x and $y = 0$

$\Rightarrow z = -4$

$Q = (0, 0, -4)$

$\vec{PQ} = [0, 0, -3]$

$$d = \frac{|[2, -1, -1] \cdot [0, 0, -3]|}{\sqrt{(2)^2 + (-1)^2 + (-1)^2}}$$

$$d = \frac{3}{\sqrt{4+1+1}} = \frac{3}{\sqrt{6}} \doteq 1.22 \text{ units}$$